

MISTY: Multidimensional Identification of Synthetic Text Yield

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Abstract—The proliferation of large language models has made distinguishing between human-written and AI-generated text a critical challenge across academic, journalistic, and regulatory domains. In this work, we propose MISTY (Multidimensional Identification of Synthetic Text Yield), a novel hybrid detection framework that integrates supervised contrastive learning with rich linguistic feature engineering to robustly identify AI-generated content. MISTY first encodes input texts using the `all-MiniLM-L6-v2` sentence transformer, then optimizes a supervised contrastive projection head to learn semantically separable embeddings. These embeddings are further fused with a comprehensive set of stylistic, syntactic, discourse, and entropy-based features—including perplexity from GPT-2, readability scores, part-of-speech distributions, and lexical entropy—yielding a 146-dimensional feature vector. A neural classifier trained on this enriched representation distinguishes human and AI text with high precision. Our results demonstrate that MISTY is able to achieve excellent outcomes. This framework offers a lightweight, extensible, and interpretable approach for the detection of synthetic text in practical applications.

Index Terms—supervised learning, natural language processing, artificial intelligence, text categorization

I. INTRODUCTION

Over the past few decades, extensive research has been carried out in various domains of artificial intelligence. Today, AI is being applied in ways that were unimaginable just a few years ago, one notable example being its use in generating text through advanced AI platforms.

Today, a wide range of AI-powered platforms and tools such as Gemini, ChatGPT, Microsoft Copilot, and QuillBot are available for generating or paraphrasing text. While these tools have undoubtedly simplified many tasks, relying too heavily on them can pose serious risks. Increasingly, students are submitting academic reports produced entirely by AI, and professionals are using these tools to draft business documents with little to no personal input. This not only diminishes the originality and authenticity of the content but also opens the door to widespread misuse, such as the creation of fraudulent documents. Being able to detect AI-generated text is becoming crucial, especially in contexts where authenticity matters. Motivated by these concerns, this research aims to address the growing need to distinguish between human-written and AI-generated content. On a publicly available dataset, we have conducted our research.

Early approaches to AI-text detection relied heavily on surface-level heuristics, perplexity thresholds, or stylistic patterns. However, with the growing sophistication of generative models, these signals have become increasingly unreliable and easy to circumvent. To address this challenge, we introduce MISTY — **M**ultidimensional **I**dentification of **S**ynthetic **T**ext **Y**ield — a hybrid detection framework that leverages deep semantic embeddings, supervised contrastive learning, and a rich set of linguistic and statistical features for robust AI-text classification.

MISTY operates in three stages: (1) it encodes text into semantic vectors using a pretrained transformer, (2) it trains a supervised contrastive projection head to separate representations of human and synthetic text in latent space, and (3) it enriches these representations with textual features such as GPT-2 perplexity, readability scores, discourse structure, part-of-speech distributions, and lexical entropy. The final classification is performed using a lightweight neural classifier trained on these multidimensional features.

This multidimensional fusion of features allows MISTY to exploit both high-level semantic structure and low-level linguistic inconsistencies introduced by language models. The use of supervised contrastive learning enables the model to maximize the separability of representations under label supervision, making it more resilient to the subtleties of modern text generators. Our results demonstrate that MISTY is capable of achieving excellent results for the categorization task.

In the following sections, we have discussed related prior experiments, our proposed methodology, and evaluation results.

II. LITERATURE REVIEW

We have studied the existing researches on AI generated text detection to gain insights which can be very helpful.

Abburi et al. [1] worked on AI generated text detection with an ensemble approach. Multiple datasets had been used for evaluation purposes. In the employed approach, each input was passed through five pre-trained models that were DeBERTa large, RoBERTa with XNLI, RoBERTa large, RoBERTa base OpenAI detector, XLM-RoBERTa NLI. For maximising the advantages of the approaches, independently generated class probabilities by each model were concatenated and passed as an input for training a voting classifier which were Logistic

Regression, Random Forest, Gaussian Naive Bayes, Support Vector Machine. The experiment achieved a highest accuracy of 78.4%.

Shah et al. [2] used a hybrid approach with stylistic features for detecting AGTs. Different aspects of Lexical features, Readability scores, Diversity and richness of the vocabulary were taken into consideration. The researchers prepared custom datasets for the study. In the experiment, at first classification was done with Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, Gradient Boosting; then, results from the top two performing classifiers were collaborated with two xAI analysis techniques which were SHAP and LIME. The hybrid approach showed promising outcomes.

Yan et al. [3] prepared a custom dataset and used RoBERTa and SVM for AGT detection. The RoBERTa model was fine tuned and the SVM model used a number of features generated by the in-house scoring engine. RoBERTa achieved 99.75% accuracy while SVM got 96%.

Mitchel et al. [4] used probability curvature for detecting machine generated texts. The researchers opinionated that LLM generated texts have negative curvature. They verified that minor rewrites of texts generated by models have lower log probability under the model compared to the original text while the opposite happens for human written texts.

Yadagiri et al. [5] focused on different linguistic features while working on detection of AGTs using pre-trained models. The researchers created a dataset of approximately forty thousand inquiries alongside responses from ChatGPT and humans. The employed models were CNN-BiLSTM, RNN, GPT-2, RoBERTa. Among the models, the best performing one was RoBERTa which achieved an accuracy of 99.73%.

Zhang et al. [6] adopted a hybrid approach for detecting AI generated texts. In the hybrid model, TF-IDF worked with Bayesian Classifier, SGD classifier, LightGBM classifier, CatBoost classifier and BPE tokenisation worked with multiple instances of Bayesian classifier. Then, this ensemble pipeline was implemented with 12 instances of DeBERTa v3 large models. The experimental approach gained the highest ROC-AUC value of 97.5%.

Qazi et al. [7] prepared a novel dataset and proposed a model for detection of GPT-generated texts in English. The researchers created GPT Reddit Dataset (GRiD) which consists of 6513 samples. 1368 were generated by GPT-3.5-turbo model while the rest were authored by humans. The human authored texts were collected from Reddit using PRAW Python and the posts dated before the official release date of the ChatGPT web application. The researchers proposed a novel method, GPTEN where tensor decomposition was leveraged for identifying underlying patterns of data. The scientists used BERT, SVM and Random Forest along with the proposed model. Out of the four, BERT outperformed others.

Gryka et al. [8] worked on AI generated email detection using different classifiers such as Decision Tree, Random Forest (100 trees), MLP classifier, KNN (K=5), SVM with radial basis function kernel, AdaBoost (50 Decision Trees as

estimators and a learning rate of 1), Gaussian Naive Bayes and Quadratic Discriminant Analysis. The dataset in the study had in total over 10k emails. The researchers worked on two languages, English and Polish. Random Forest classifier outperformed other models with 98.8% F1 score for English and 92.1% for Polish.

Zhou and Wang [9] worked on detection of AI generated texts in cross domains. Combining multiple datasets, a dataset was prepared for the study. The researchers used RoBERTa-Ranker, a modified version of RoBERTa where the margin ranking loss function was used and a mean pooling layer was added to the encoding layer.

Guo et al. [10] worked on a pipeline for AI generated text detection which combined a multi-level contrastive learning technique with multi task learning. Firstly, data from training samples were pre-encoded and features were extracted. For any text, similarity between encoded features and each feature vector nestled in the feature vector was calculated then. Finally, for prediction KNN had been employed.

Wu et al. [11] experimented with various detectors for LLM generated text detection. For the dataset, human written texts were collected from existing datasets and platforms while different LLMs such as GPT-3.5-turbo, PaLM-2-bison, Claude-instant, Llama-2-70b were employed for LLM generated texts.

Guo et al. [12] opinionated in their study that casual language models, when encountering human data, memorize more proceeding token information while predicting less next token information in their output logits. The researchers used a bi-directional calculation method measuring cross entropy losses between an output logits and the ground truth forward token and between the output logits and the immediately preceding input backward token.

Wang et al. [13] made use of GBERT for AI generated text detection. They used a private dataset with 670 human written texts and 708 ai generated ones. The experiment achieved 98.1% training accuracy and 97.71% testing accuracy on average.

Yu et al. [14] worked on creating a dataset for detection of Chat-GPT generated abstracts. The dataset had 15395 human written abstracts and 35304 were ChatGPT generated. The human written abstracts were collected from IEEE Xplore based on specific keywords. The ChatGPT generated ones were categorized into three kinds which are Generation, Polish and Mix. In Generation, abstracts were generated completely by ChatGPT; as for Polish, a human written text was given as input and it was polished by the tool; in Mix, abstracts consisted of both ChatGPT polished and human written. On the prepared dataset, a number of BERT variants were applied and highest AUC scores were for Polish, 99.56; for Generation, 100 and for Mix, 87.83.

Pudasaini et al. [15] worked with ArguGPT, RADAR (Robust AI text Detector via Adversarial Learning) and OpenAI Detector in regards to AI generated text identification. M4 Dataset and HC3 Dataset were used in the study.

McGovern et al. [16] opinionated that with appropriate features, it is possible to detect AI generated texts using simple

classifiers. The researchers used a GradientBoost classifier along with three feature sets which were word n-grams, character n-grams and parts-of-speech n-grams. In the study a number of datasets were used and the proposed methodology showed promising results.

Varadarajan et al. [17] expressed that LLM generated texts have limited humanness and diversity in them. They opined that human factors such as demographics, personality, empathy and behavioral linguistic traits are of great significance. Multiple datasets were employed in the study. In the experiment, unsupervised classification was done using spectral clustering with radial basis function.

Andric et al. [18] used a classification method that combined graph representation of texts with GNN for AI generated text detection. Different node feature initialization strategies had been considered in the study. A Co-Occurrence graph had been employed and fine tuning was done with pre-trained models such as BERT-Base-Uncased or RoBERTa-Base. The processed graph was fed to GNN with a GAT layer. The graph document embedding was fed into a classifier, for which a dense neural network had been implemented. The Autexification2023 dataset was used for the study.

Abbas [19] worked on AI generated text identification utilising a zero shot prompt along with Sentence BERT (SBERT). Also, graph convolutional network model was integrated which enhanced the accuracy. The PAN-AP-2019 dataset had been used in the experiment.

The researchers in a study [20] worked with multiple ML and DL classification models for AI-generated text detection. They created a dataset with 509 descriptive question-answers where the answers were prepared by both humans and the Chat-GPT engine. In the study, the RoBERTa-based model outperformed others.

Alhijawi et al. [21] created a dataset of 3000 records for LLM generated content detection. The records were of three types - Human written, AI generated, Mixed and each type had 1000 records. The researchers experimented with various approaches and presented a hybrid model of MLP and CNN which achieved a highest accuracy of 0.876 in the experiment.

III. DATASET

We have used a publicly available dataset from Kaggle [22] uploaded by Darek Kłeczek [23] for AI Generated text detection. The dataset has 44868 text records and 27371 records are human written while the rest are AI generated. The AI generated texts are labeled as 1 while texts written by humans are labeled as 0. The dataset is available in CSV file format.

IV. PROPOSED METHODOLOGY

We propose MISTY, a hybrid detection framework that leverages contrastive learning and a multidimensional feature space to accurately classify AI-generated vs. human-written text. The architecture consists of the following key components:

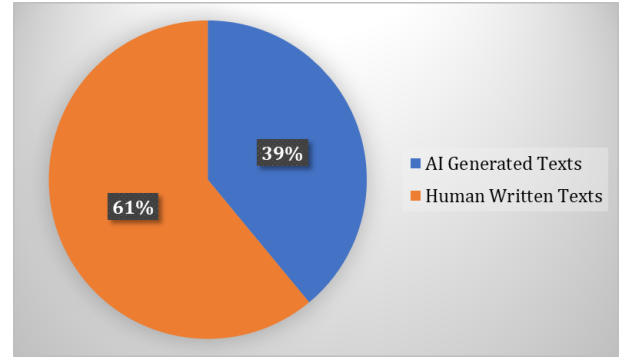


Fig. 1. Percentage of Category wise Records in Dataset

A. Input and Embedding Layer

Raw text inputs are first transformed into fixed-size dense vectors using the all-MiniLM-L6-v2 [24] sentence transformer. These embeddings serve as the semantic foundation for both contrastive learning and classification.

B. Supervised Contrastive Learning

A projection head comprising a two-layer feedforward network maps the embeddings to a latent contrastive space. Using labeled data, we optimize a Supervised Contrastive Loss, where samples from the same class are pulled together, and samples from different classes are pushed apart. This fine-tunes the representation space for downstream classification.

C. Linguistic Feature Extraction

To enhance detection accuracy beyond semantic representations, MISTY integrates a rich suite of linguistic features:

- Perplexity: Estimated using a pre-trained GPT-2 model to quantify the language modeling fluency.
- Stylometric Features: Readability scores (e.g., Flesch, Dale-Chall, ARI) computed via textstat to model writing complexity.
- Discourse Features: Captures sentence length variance and discourse connectives (e.g., “however”, “moreover”) as proxies for coherence.
- POS Frequency Features: Frequencies of specific part-of-speech tags (NN, VB, JJ, RB) obtained from NLTK POS tagging.
- Token Entropy: Shannon entropy over word frequencies to measure lexical richness and repetition.

These features are concatenated with the contrastively projected embedding to form a unified 146-dimensional input vector.

D. Classification Layer

The final classifier is a two-layer neural network with dropout regularization. It maps the fused feature vector to binary outputs indicating whether the input is AI-generated or human-authored.

E. Training Pipeline

The contrastive projection head is first trained independently using supervised pairs. Then, the classifier is trained on concatenated vectors combining contrastive representations with handcrafted linguistic features. Both training phases use the Adam optimizer with a moderate learning rate ($1e-3$), and losses are monitored over multiple epochs for convergence.

F. Prediction Module

Given a new input, MISTY computes all associated features and returns the predicted label along with the classification confidence. This system generalizes well due to its hybrid nature—integrating semantic similarity, structural analysis, and stylistic depth.

V. EVALUATION RESULTS

For evaluation of the text classification task, we have used Accuracy, Precision, Recall and F1 score metrics. The outcomes of our ensemble model is illustrated in the Fig 3.

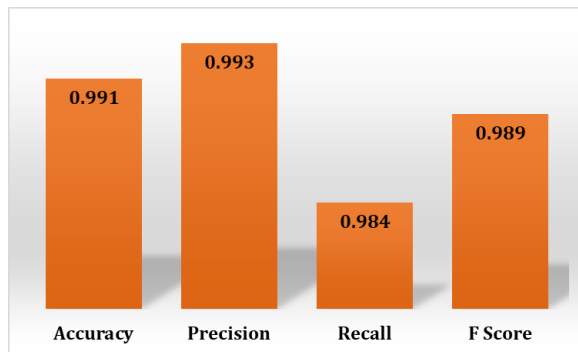


Fig. 2. Performance of the Proposed Methodology

We observe that the proposed method achieves excellent outcomes.

The model achieves excellent performance due to the synergy of several powerful innovations in its architecture and training methodology.

- The model uses Supervised Contrastive Learning (SupCon) to optimize the separation between human-written and AI-generated sentences in the embedding space. Unlike cross-entropy loss, SupCon encourages samples of the same class to cluster and pushes dissimilar ones apart, which leads to more discriminative representations for downstream classification.
- Instead of directly using raw sentence embeddings, a learnable projection head maps embeddings to a latent space optimized for contrastive separation. This improves inter-class margin and reduces representation overlap, especially between subtly different text sources.
- The final classifier does not rely solely on embeddings; it incorporates a diverse set of handcrafted features such as Perplexity, Stylometry Metrics, Discourse Features, POS Frequencies, Lexical Entropy. These features are orthogonal to embeddings, allowing the model to capture

both deep semantic similarity and surface-level statistical artifacts.

- The deep neural classifier offers nonlinear feature interactions and joint optimization of all input features. The dropout layers and ReLU activations improve generalization and mitigate overfitting on small samples.
- Both the projection head and final classifier are trained with supervised signals, allowing full end-to-end optimization for classification.

The system was tested on a standard CUDA-enabled machine. Inference time per sample, including embedding and feature extraction, was under 500 ms on average, making the system viable for near-real-time applications.

VI. THE AVAILABILITY OF THE RESOURCES

The resources related to the study will be publicly available on a GitHub repository [MdSiamAnsary/ClassificationTasks](https://github.com/MdSiamAnsary/ClassificationTasks) [25].

VII. CONCLUSION AND FUTURE WORKS

MISTY is a new pipeline that combines contrastive learning with a dense span of linguistic and statistical features, detecting AI-generated text. The system uses fine-tuned sentence embedding using supervised contrastive loss, including perplexity scores using GPT-2 and stylometric indices, discourse-level features and part-of-speech statistics, and lexical entropy. We substituted traditional classifiers with a light deep neural architecture to achieve improved results. In our findings, we mention the significance of the encoding of semantic representations and the encoding of handcrafted semantic interpretability to obtain both surface and latent variations in the style of text generation. The addition of discourse and entropy measures led to a greater classification accuracy with especially the syntactically simple but semantically manipulated texts. We would like to evaluate it on a large scale (with large corpus and diverse corpora) in the future as a part of work, with real-world output of generative content. We are also planning to include structural and rhetorical indications, e.g. discourse parsing trees or coherence chains but extend the model to multilingual settings and test cross-lingual transferability. and develop an explainability module to visualize the impact of which features contribute most to the decision of the classification. With the unceasing development of MISTY in the context of deeper knowledge and a wider linguistic scope, we expect to contribute to more reliable AI text verification systems in academic, journalistic and forensic fields.

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